





section 9

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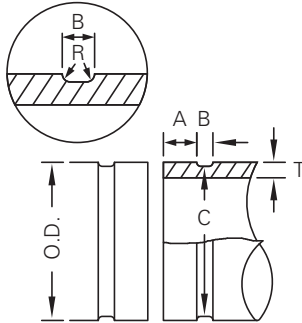
AWWA ductile iron series



Shurjoint offers a variety of grooved mechanical couplings and fittings for AWWA ductile iron pipe in sizes 3" to 12". The Shurjoint coupling features a two-piece housing and GapSeal gasket for a leak-tight seal. Ductile iron pipe shall be cut-grooved to AWWA C606 Table 2 and Table 3 - Radius Cut Groove Specifications.

Rubber gaskets are specially compounded to seal on ductile iron surfaces and are available in three grades to meet your service requirement needs. See page 231 for details.

radius cut grooved dimensions - ductile iron pipe



gasket seating surface (A):

The same coupling can be used either as a rigid joint or a flexible joint depending on the groove. Gasket seat "A Rigid" is for rigid joints and Gasket seat "A Flex." for flexible joints.

The gasket seating surface shall be free from deep scores, marks, or ridges that could prevent a positive seal.

groove diameter (C):

The "C" diameters are average values. The groove must be of uniform depth around the entire pipe circumference.

radius (R):

The groove must be cut with a radius 'R' at the corners of the groove to reduce stress concentration.

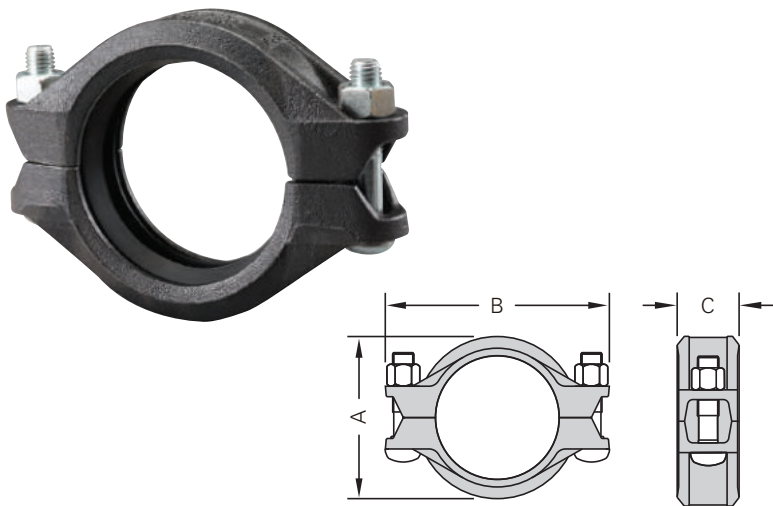
minimum wall thickness (T):

"T" is the minimum allowable wall thickness that may be cut-grooved; tolerances are to conform to ANSI/AWWA C151/A21.51.

AWWA ductile iron pipe

nom. size	pipe O.D.		gasket seat A				groove dia. C			
	basic	tolerance		rigid +0/-0.02 +0/-0.51	flex. +0.016/-0.047 +0.41/-1.19	groove width B +0.031/-0.016 +0.79/-0.41	basic	tol. +0 +0	radius R	min. allowed wall thickness T
		+	-							
3"	3.96"	+0.045"	-0.045"	0.840"	0.750"	0.375"	3.723"	-0.020"	0.120"	0.31"
80	(100.6 mm)	(+1.14 mm)	(-1.14 mm)	(21.34 mm)	(19.05 mm)	(9.53 mm)	(94.56 mm)	(-0.51 mm)	(3.05 mm)	(7.9 mm)
4"	4.80"	+0.06"	-0.06"	0.840"	0.750"	0.375"	4.563"	-0.020"	0.120"	0.32
100	(121.9 mm)	(+1.52 mm)	(-1.52 mm)	(21.34 mm)	(19.05 mm)	(9.53 mm)	(115.90 mm)	(-0.51 mm)	(3.05 mm)	(8.1 mm)
6"	6.90"	+0.06"	-0.06"	0.840"	0.750"	0.375"	6.656"	-0.020"	0.120"	0.34
150	(175.3 mm)	(+1.52 mm)	(-1.52 mm)	(21.34 mm)	(19.05 mm)	(9.53 mm)	(169.06 mm)	(-0.51 mm)	(3.05 mm)	(8.6 mm)
8"	9.05"	+0.06"	-0.06"	0.950"	0.875"	0.500"	8.781"	-0.025"	0.145"	0.36
200	(229.9 mm)	(+1.52 mm)	(-1.52 mm)	(24.13 mm)	(22.23 mm)	(12.70 mm)	(223.04 mm)	(-0.64 mm)	(3.68 mm)	(9.1 mm)
10"	11.10"	+0.06"	-0.06"	1.015"	0.938"	0.500"	10.813"	-0.025"	0.145"	0.38
250	(281.9 mm)	(+1.52 mm)	(-1.52 mm)	(25.78 mm)	(23.83 mm)	(12.70 mm)	(274.65 mm)	(-0.64 mm)	(3.68 mm)	(9.7 mm)
12"	13.20"	+0.06"	-0.06"	1.015"	0.938"	0.500"	12.906"	-0.030"	0.145"	0.40
300	(335.3 mm)	(+1.52 mm)	(-1.52 mm)	(25.78 mm)	(23.83 mm)	(12.70 mm)	(327.81 mm)	(-0.76 mm)	(3.68 mm)	(10.2 mm)

A505 coupling



Shurjoint Model A505 Couplings are designed for connecting grooved ductile iron pipe and fittings of ANSI/AWWA C151/ A21.51, Class 53 or higher dimensions. The same coupling can be used either as a flexible coupling or a rigid coupling depending on the type of groove processed. The Model A505 Coupling is recommended for service up to 500 psi (35 Bar) depending on the size. The GapSeal gasket fits flush over the pipe ends and prevents fluids from entering into the gasket cavity.

dimensions



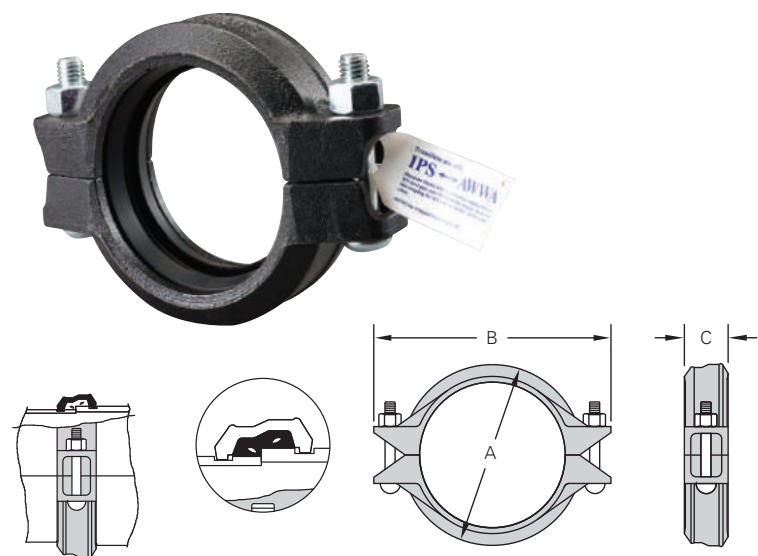
AWWA D.I pipe					bolt		dimensions			weight
nom. size	pipe O.D.	max. working pressure (CWP)*	max. end load	axial displacement	no.	size	A	B	C	
3"	3.96"	500 psi	6200 lbs	0-0.09"	2	1/2 x 3"	5.33"	7.36"	2.40"	5.5 lbs
80	(100.6 mm)	(35 bar)	(27.59 kN)	(0-2.4 mm)			(136 mm)	(187 mm)	(61 mm)	(2.5 Kgs)
4"	4.80"	500 psi	9000 lbs	0-0.09"	2	5/8 x 3 1/2"	6.96"	8.33"	2.32"	7.3 lbs
100	(121.9 mm)	(35 bar)	(40.50 kN)	(0-2.4 mm)			(157 mm)	(212 mm)	(59 mm)	(3.3 Kgs)
6"	6.90"	400 psi	14950 lbs	0-0.09"	2	5/8 x 3 1/2"	9.37"	10.75"	2.44"	9.7 lbs
150	(175.3 mm)	(28 bar)	(66.26 kN)	(0-2.4 mm)			(211 mm)	(273 mm)	(62 mm)	(4.4 Kgs)
8"	9.05"	400 psi	25600 lbs	0-0.16"	2	3/4 x 4 3/4"	12.07"	13.94"	2.64"	18.7 lbs
200	(229.9 mm)	(28 bar)	(113.92 kN)	(0-4.0 mm)			(272 mm)	(354 mm)	(67 mm)	(8.5 Kgs)
10"	11.10"	350 psi	33850 lbs	0-0.16"	2	3/4 x 4 3/4"	14.50"	16.06"	2.95"	24.2 lbs
250	(281.9 mm)	(24 bar)	(150.63 kN)	(0-4.0 mm)			(326 mm)	(408 mm)	(75 mm)	(11.0 Kgs)
12"	13.20"	350 psi	47900 lbs	0-0.16"	2	3/8 x 6 1/2"	16.89"	18.19"	2.95"	30.8 lbs
300	(335.3 mm)	(24 bar)	(211.50 kN)	(0-4.0 mm)			(380 mm)	(462 mm)	(75 mm)	(14.0 Kgs)

* Pressure ratings listed are based on radius cut-grooved Thickness Class 53 or higher pipe.
† Only when connected on flexible radius groove.

special gaskets for AWWA ductile iron pipe

grade	temp. range	compound	color code	general service recommendations
S	-20°F to +180°F (-29°C + 82°C)	nitrile	 red stripe	Specially formulated to seal on ductile iron pipe iron surfaces. Recommended for petroleum products, air with oil vapors, vegetable and mineral oils within the specified temperature range; except hot dry air over +140°F (60°C) and water over +150°F (65°C) Not Recommended for Hot Water Services.
M	-20°F to +200°F (-29°C + 93°C)	halogenated butyl	 brown stripe	Specially formulated to seal on ductile iron pipe surfaces. Recommended for water service plus a variety of dilute acids, oil-free air and other chemical services within the specified temperature range. The compound is UL classified per NSF/ANSI 61 and NSF/ANSI 372 for potable water applications. Not Recommended for Hot Water Services.
L	-30°F to +350°F (-34°C + 177°C)	silicone	 red gasket	Recommended for dry heat, air without hydrocarbons to 350°F (177°C) and certain chemical services.

A507 transition coupling



The Shurjoint Model A507 Transition Coupling provides for a direct connection between grooved end IPS steel pipe and grooved end AWWA ductile iron pipe, fittings and or valves. The A507 will accommodate roll or cut grooved IPS steel pipe and rigid or flexible AWWA ductile iron cut grooves.

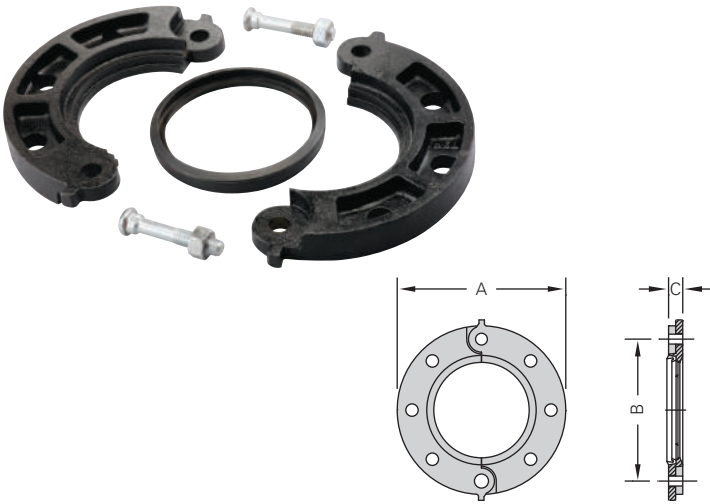
dimensions

nom. size	pipe O.D.		max. working pressure (CWP)*	max. end load	axil displacement	bolt size	dimensions			weight
	IPS steel	AWWA ductile iron					A	B	C	
3"	3.500"	3.96"	500 psi	4810 lbs	0.03"	1/2 x 2 3/4"	5.31"	7.28"	2.13"	4.8 lbs
80	(88.9 mm)	(100.6 mm)	(35 bar)	(21.41 kN)	(1 mm)		(135 mm)	(185 mm)	(54 mm)	(2.2 Kgs)
4"	4.500"	4.80"	500 psi	7950 lbs	0.03"	5/8 x 3 1/2"	6.22"	8.54"	2.19"	6.4 lbs
100	(114.3 mm)	(121.9 mm)	(35 bar)	(35.38 kN)	(1 mm)		(158 mm)	(217 mm)	(56 mm)	(2.9 Kgs)
6"	6.625"	6.90"	400 psi	13780 lbs	0.03"	5/8 x 3 1/2"	8.23"	10.75"	2.19"	8.6 lbs
150	(168.3 mm)	(175.3 mm)	(28 bar)	(61.32 kN)	(1 mm)		(209 mm)	(273 mm)	(56 mm)	(3.9 Kgs)
8"	8.625"	9.05"	400 psi	23370 lbs	0.03"	3/4 x 4 3/4"	10.79"	13.66"	2.52"	16.7 lbs
200	(219.1 mm)	(229.9 mm)	(28 bar)	(103.99 kN)	(1 mm)		(274 mm)	(347 mm)	(64 mm)	(7.6 Kgs)
10"	10.750"	11.10"	350 psi	33850 lbs	0.03"	7/8 x 6 1/2"	13.03"	16.10"	2.76"	24.9 lbs
250	(273.0 mm)	(281.9 mm)	(24 bar)	(155.96 kN)	(1 mm)		(331 mm)	(409 mm)	(70 mm)	(11.3 Kgs)
12"	12.750"	13.20"	350 psi	47870 lbs	0.03"	7/8 x 6 1/2"	15.00"	18.35"	2.76"	28.6 lbs
300	(323.9 mm)	(335.3 mm)	(24 bar)	(220.64 kN)	(1 mm)		(381 mm)	(466 mm)	(70 mm)	(13.0 Kgs)

* Pressure ratings listed are based on radius cut-grooved Thickness Class 53 or higher pipe



A512 flange adapter



The Shurjoint Model A512 Flange Adapter provides for the direct connection between AWWA ductile iron radius pipe grooves and flanged components. The two part flange features integral closure tabs to aid in assembly. Note: As with other flange adapters the A512 requires a sufficient smooth flat mating area for proper sealing, please refer to the A512 cut sheet or contact Shurjoint for details.

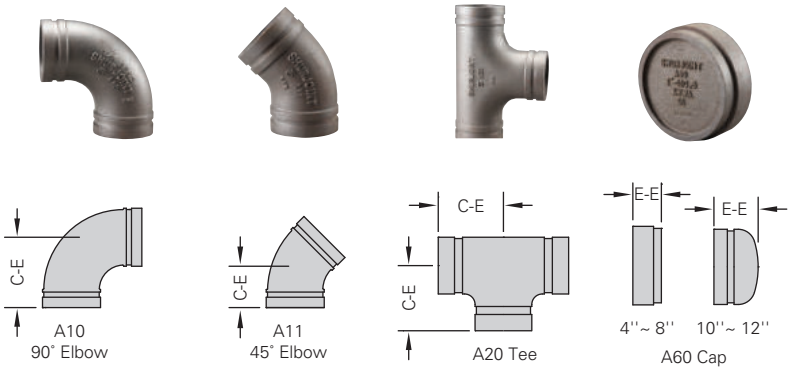
dimensions

AWWA D.I pipe				bolt		dimensions			weight
nom. size	pipe O.D.	max. working pressure (CWP)*	max. end load	no.	size	A	B	C	
3"	3.96"	250 psi	3100 lbs	4	5/8 x 3"	7.50"	6.00"	1.10"	8.6 lbs
80	(100.6 mm)	(17 bar)	(13.80 kN)			(190 mm)	(152 mm)	(28 mm)	(3.9 Kgs)
4"	4.80"	250 psi	4500 lbs	8	5/8 x 3"	9.00"	7.50"	1.10"	9.9 lbs
100	(121.9 mm)	(17 bar)	(20.03 kN)			(229 mm)	(191 mm)	(28 mm)	(4.5 Kgs)
6"	6.90"	250 psi	9300 lbs	8	3/4 x 3 1/2"	11.00"	9.49"	1.10"	12.0 lbs
150	(175.3 mm)	(17 bar)	(41.39 kN)			(279 mm)	(241 mm)	(28 mm)	(5.4 Kgs)
8"	9.05"	250 psi	16000 lbs	8	3/4 x 3 1/2"	13.50"	11.75"	1.22"	18.9 lbs
200	(229.9 mm)	(17 bar)	(71.20 kN)			(343 mm)	(298 mm)	(31 mm)	(8.6 Kgs)
10"	11.10"	250 psi	23700 lbs	12	7/8 x 4"	16.00"	14.25"	1.26"	25.3 lbs
250	(281.9 mm)	(17 bar)	(105.47 kN)			(406 mm)	(362 mm)	(32 mm)	(11.5 Kgs)
12"	13.20"	250 psi	34000 lbs	12	7/8 x 4"	19.00"	17.00"	1.26"	34.4 lbs
300	(335.3mm)	(17 bar)	(151.30 kN)			(483mm)	(432mm)	(32 mm)	(15.6Kgs)

* Pressure ratings listed are based on radius cut-grooved Thickness Class 53 or higher pipe.

** Flange drilling to ANSI B16.1 Class 125.

- A10 90° elbow
- A11 45° elbow
- A20 tee
- A60 cap

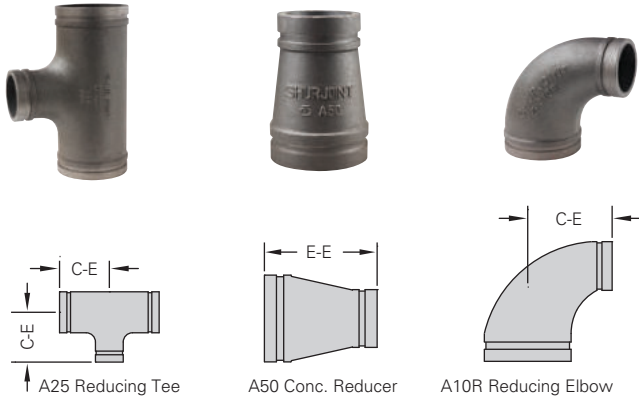


Shurjoint AWWA grooved end fittings are supplied with rigid radius grooves as per ANSI / AWWA C606. The fittings also conform to ANSI A21.10 / AWWA C110 for center to end (C to E) dimensions and AWWA C153 or ANSI A21.10 / AWWA C110 for wall thickness. Pressure ratings are 500 psi (35 Bar). Fittings are supplied painted black. Other surface finish options including 'non-coated' and thermoplastic coating are available on request.

dimensions

AWWA D.I pipe		A10 90° elbow		A10 45° elbow		A20 tee		A60 cap	
nom. size	pipe O.D.	C - E	weight	C - E	weight	C - E	weight	E - E	weight
3"	3.96"	5.50"	9.2 lbs	3.00"	5.8 lbs	5.50"	13.4 lbs	1.22"	2.6 lbs
80	(100.6 mm)	(140 mm)	(4.2 Kgs)	(76 mm)	(2.6 Kgs)	(140 mm)	(6.1 Kgs)	(31 mm)	(1.2 Kgs)
4"	4.80"	6.50"	12.3 lbs	4.00"	9.2 lbs	6.50"	18.5 lbs	1.22"	3.5 lbs
100	(121.9 mm)	(165 mm)	(5.6 Kgs)	(102 mm)	(4.2 Kgs)	(165 mm)	(8.4 Kgs)	(31 mm)	(1.6 Kgs)
6"	6.90"	8.00"	23.5 lbs	5.00"	17.6 lbs	8.00"	35.0 lbs	1.22"	6.8 lbs
150	(175.3 mm)	(203 mm)	(10.7 Kgs)	(127 mm)	(8.0 Kgs)	(203 mm)	(15.9 Kgs)	(31 mm)	(3.1 Kgs)
8"	9.05"	9.00"	41.6 lbs	5.50"	33.0 lbs	9.00"	60.3 lbs	1.46"	13.0 lbs
200	(229.9 mm)	(229 mm)	(18.9 Kgs)	(140 mm)	(15.0 Kgs)	(229 mm)	(27.4 Kgs)	(37 mm)	(5.9 Kgs)
10"	11.10"	11.00"	64.9 lbs	6.50"	44.9 lbs	11.00"	110.5 lbs	2.99"	22.0 lbs
250	(281.9 mm)	(279 mm)	(29.5 Kgs)	(165 mm)	(20.4 Kgs)	(279 mm)	(45.7 Kgs)	(76 mm)	(10.0 Kgs)
12"	13.20"	12.00"	94.2 lbs	7.50"	72.0 lbs	12.00"	136.0 lbs	2.99"	24.6 lbs
300	(335.3 mm)	(305 mm)	(42.8 Kgs)	(191 mm)	(32.7 Kgs)	(305 mm)	(61.7 Kgs)	(76 mm)	(11.2 Kgs)

A25 reducing tee
A50 concentric reducer
A10R reducing elbow



Shurjoint AWWA grooved end fittings are supplied with rigid radius grooves as per ANSI / AWWA C606. The fittings also conform to ANSI A21.10 / AWWA C110 for center to end (C to E) dimensions and AWWA C153 or ANSI A21.10 / AWWA C110 for wall thickness. Pressure ratings are 500 psi (35 Bar). Fittings are supplied painted black. Other surface finish options including 'non-coated' and thermoplastic coating are available on request.

dimensions

AWWA D.I pipe		A25 reducing tee		A50 concentric reducer		A10R 90° reducing elbow	
nom. size	O.D.	C - E	weight	E - E	weight	C - E	weight
4 x 3"	4.80 x 3.96"	6.50"	18.8 lbs	7.00"	7.7 lbs	6.50"	12.1 lbs
100 x 80	(121.9 x 100.6 mm)	(165 mm)	(8.5 Kgs)	(178 mm)	(3.5 Kgs)	(165 mm)	(5.5 Kgs)
6 x 4"	6.90 x 4.80"	8.00"	33.3 lbs	9.00"	14.3 lbs	8.00"	25.1 lbs
150 x 100	(175.3 x 121.9 mm)	(203 mm)	(15.2 Kgs)	(229 mm)	(6.5 Kgs)	(203 mm)	(11.4 Kgs)
8 x 4"	6.90 x 4.80"	9.00"	51.2 lbs	11.00"	24.0 lbs	9.00"	43.5 lbs
200 x 100	(175.3 x 121.9 mm)	(229 mm)	(23.3 Kgs)	(279 mm)	(10.9 Kgs)	(229 mm)	(19.8 Kgs)
8 x 6"	9.05 x 6.90"	9.00"	57.2 lbs	11.00"	24.0 lbs	9.00"	41.8 lbs
200 x 150	(229.9 x 175.3 mm)	(229 mm)	(26.0 Kgs)	(279 mm)	(12.6 Kgs)	(229 mm)	(19.0 Kgs)
10 x 4"	11.10 x 4.80"	11.00" *	120.0 lbs	12.00" *	42.0 lbs	-	-
250 x 100	(281.9 x 121.9 mm)	(279 mm)	(54.4 Kgs)	(305 mm)	(19.1 Kgs)	-	-
10 x 6"	11.10 x 4.80"	11.00" *	128.0 lbs	12.00"	46.0 lbs	11.00"	77.0 lbs
250 x 150	(281.9 x 121.9 mm)	(279 mm)	(58.1 Kgs)	(305 mm)	(20.9 Kgs)	(279 mm)	(34.9 Kgs)
10 x 8"	11.10 x 4.80"	11.00" *	130.0 lbs	12.00"	50.0 lbs	11.00"	88.0 lbs
250 x 200	(281.9 x 121.9 Kgs)	(279 mm)	(59.0 Kgs)	(305 mm)	(22.7 Kgs)	(279 mm)	(39.9 Kgs)
12 x 4"	13.20 x 4.80"	12.00" *	112.0 lbs	14.00" *	60.0 lbs	-	-
300 x 100	(335.3 x 121.9 mm)	(305 mm)	(50.8 Kgs)	(356 mm)	(27.2 Kgs)	-	-
12 x 6"	13.20 x 4.80"	12.00" *	180.0 lbs	14.00" *	70.0 lbs	12.00" *	110.0 lbs
300 x 150	(335.3 x 121.9 mm)	(305 mm)	(81.7 Kgs)	(356 mm)	(31.8 Kgs)	(305 mm)	(49.9 Kgs)
12 x 8"	13.20 x 4.80"	12.00" *	186.0 lbs	14.00" *	74.0 lbs	12.00" *	126.0 lbs
300 x 200	(335.3 x 121.9 mm)	(305 mm)	(84.4 Kgs)	(356 mm)	(33.6 Kgs)	(305 mm)	(57.2 Kgs)
12 x 10"	13.20 x 4.80"	12.00" *	192.0 lbs	14.00"	84.0 lbs	12.00" *	150.0 lbs
300 x 250	(335.3 x 121.9 mm)	(305 mm)	(87.1 Kgs)	(356 mm)	(38.1 Kgs)	(305 mm)	(68.0 Kgs)

* Non-standard/stock items may require longer lead time.